# João Natividade

João da Cruz de Natividade e Silva Neto (João Silva Neto) · Belém, Brazil · Remote

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## PROFESSIONAL SUMMARY

Software engineer and Computer Engineering senior at UFPA specializing in backend development, microservices, MCP automation, AWS cloud, and IoT. Co-author of registered software intellectual property at INPI (#BR512025004902-1). Hands-on experience optimizing Python APIs (FastAPI) and Rust systems, resolving mobile memory leaks (Flutter/Dart), building Dash dashboards, and crafting 120 FPS Canvas graphics.

## PROFESSIONAL EXPERIENCE

**Software Engineering Intern · LaPS / UFPA (DMóvel Assistive Technology)**

March 2026 — September 2026 | Belém, Brazil | 360 Hours

- Registered Software Intellectual Property co-author at INPI (Process #BR512025004902-1 · DMóvel · UFOPA / UFPA / UEMA · 50-year protection under Law 9.609/1998).
- Maintained and refactored Python/FastAPI backend integrated with databases and cloud services on Amazon Web Services (AWS).
- Developed and maintained cross-platform assistive UI in Flutter/Dart and interactive dashboards using Dash (Python), HTML, CSS, and JavaScript.
- Diagnosed and resolved mobile memory leaks and runtime bottlenecks in production.
- Graded 10.0/10 (Concept Excellent) by the permanent lab advisor.

## SELECTED PROJECTS

**AetherSense (Rust, ESP32, React, Python, ONNX)**

An experimental platform connecting Wi-Fi signals, processing and a web interface to explore camera-free sensing.

- Link: joaosnet.github.io/en/projects/aethersense/ | Código: github.com/joaosnet/aethersense

**DMóvel (Flutter, Dart, FastAPI, AWS, Dash)**

Assistive technology ecosystem featuring a Flutter mobile app, an async FastAPI backend on AWS, and analytical telemetry dashboards. Registered software IP co-author at INPI (#BR512025004902-1).

- Link: lapshub.github.io/dmovel_web_app/ | Código: github.com/LAPSHub/dmovel_web_app

**Varedura (Python, AsyncIO, Docker, MCP, Segurança)**

Fast CLI tool and MCP connector for subdomain enumeration, network probing, and attack-surface discovery.

- Link: joaosnet.github.io/en/projects/varedura/ | Código: github.com/joaosnet/Varedura

**Evolutionary Lab (JavaScript, Plotly, AG, PSO, ED)**

A browser-based lab comparing genetic algorithms, particle swarm optimization and differential evolution on benchmark functions.

- Link: joaosnet.github.io/evolutionary-optimization-viz-ag-pso/ | Código: github.com/joaosnet/evolutionary-optimization-viz-ag-pso

## TECHNICAL SKILLS

**Languages:** Python, Rust, JavaScript, TypeScript, Dart, C, SQL, HTML5, CSS3

**Backend & Cloud:** FastAPI, AWS, Docker, REST APIs, MCP (Model Context Protocol), Dash

**Systems & Tools:** Git, GitHub Actions, Linux, ESP32, Canvas 2D, Three.js, WebSockets

## EDUCATION

**Bachelor of Science in Computer Engineering — Universidade Federal do Pará (UFPA)**

Bachelor's degree in progress · Expected graduation: 2026 | Belém, Brazil

- Academic Honors: Maximum grade (100/100) in Hardware Design, Software Engineering, Databases, and Cloud Computing.
- Languages: Portuguese (Native) | English (Advanced) | Spanish (Professional)